



DATABASE SYSTEMS

Analysis and Design Phase

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Title of Project:

Smart Quiz Conduction System

Chapter One:

Business requirements and rules for Quiz conducting System

Business Requirements:

1. User Management:

- The system must support two types of users: **Admins** and **Students**.
- Both users must log in using a unique **username** and secure **password**.
- Admins should be able to **add**, **manage**, and **authenticate** students into the system.
- The system should ensure that **duplicate usernames** are not allowed, and passwords must be **stored securely**.
- Unauthorized access must be restricted using proper authentication mechanisms.

2. Exam Management:

- Admins must be able to **create new exams** by entering a title and description.
- Each exam can have **multiple MCQ** based questions.
- Every question must include:
 - A **question statement** (text).

- Four answer options (A, B, C, D).
- Only **one correct** answer must be specified for each question.
- Admins should be able to **edit** or **delete** existing questions.
- Each question should be associated with the **exam** it belongs to and include the **date added**.

3. Quiz Participation:

- Students should be able to **view the list** of exams available to them.
- Students should be able to **attempt the quiz** by selecting answers from multiple choices.
- The system must **display one question** at a time with clear navigation.
- Once the quiz is completed, students should **not** be able to **attempt** it again (unless allowed).

4. Result and Score Management:

- Upon quiz submission, the system must automatically **evaluate the answers**.
- The score should be calculated and **displayed to the student** immediately.
- Results should include:
 - **Total score**
 - **Percentage**
 - Optionally, a **question-by-question breakdown** of correct and incorrect answers.
- Admins must be able to **view and export reports** of students' performances.

5. Exam Scheduling (Set_Exam):

- Admins should be able to **schedule exams** for specific students.
- The system should store the **exam date** and track which students attempted which exam.

6. Data Consistency and Security:

- The system must prevent:
 - Multiple answers for a single question.
 - Submission of incomplete quizzes.
- All sensitive data (especially passwords) must **not be visible** to anyone.

Business Rules:

- Unique login credentials for each user
- Only Admin can manage exams and questions
- Students can't create or modify exams
- Proper error messages for invalid actions
- One answer per question restriction
- Submission handling for unanswered questions

List of Final Entities :

- Person
- Admin
- Student
- Exams

- Questions
- Set_Exam
- Score

List of Final Relationships

- **Students** take **Exams** (Many-to-Many):

One student can take many exams, and one exam can be taken by many students.

- **Admins** create **Exams** (One-to-Many):

One admin can create many exams, and one exam is created by one admin.

- **Exams** contain **Questions** (One-to-Many):

One exam can contain many questions, and one question belongs to one exam.

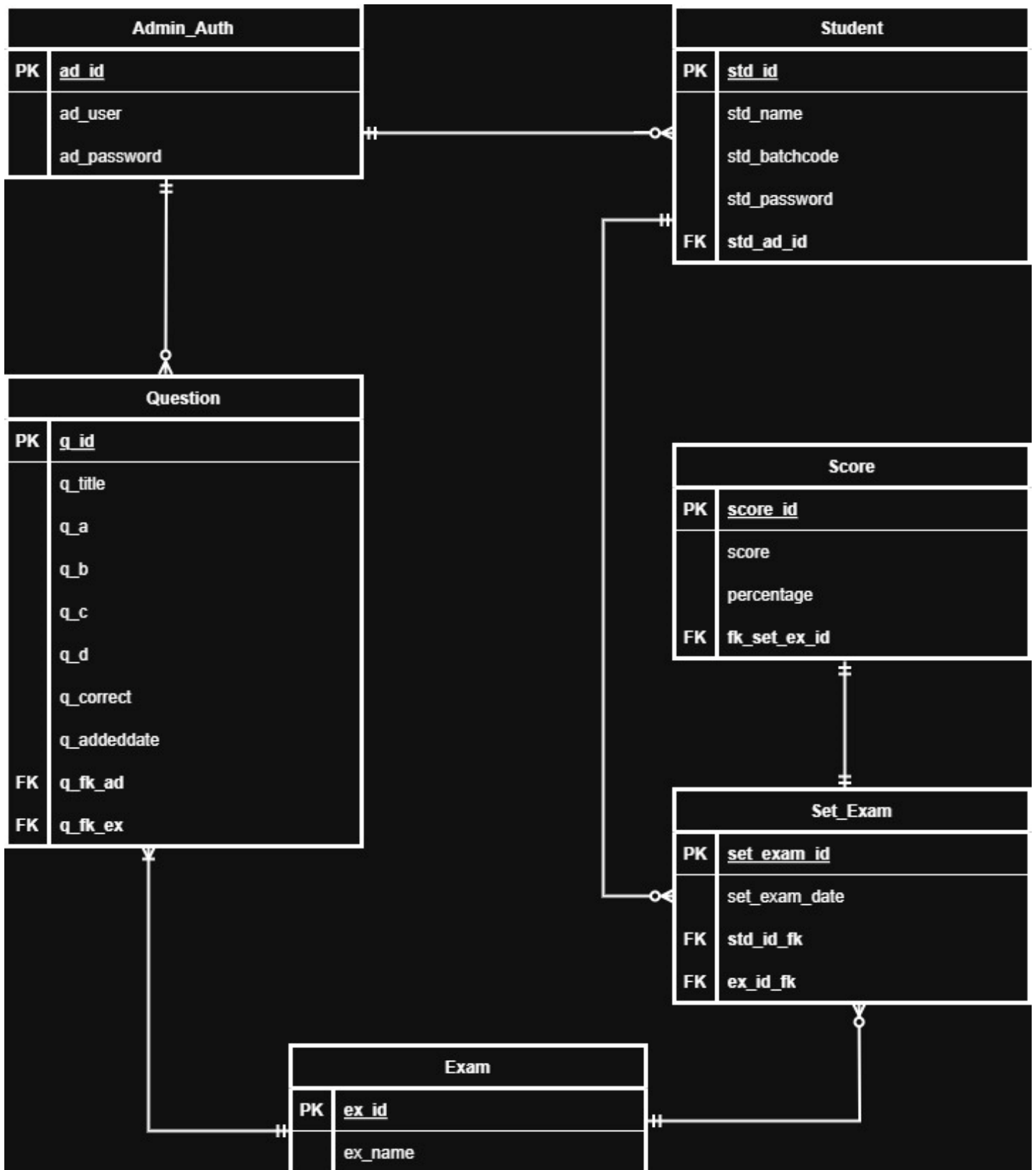
- Each **Exam** has a **Score**

- **Admin** adds **Students**

- **Admin** adds **Questions**

- A **Person** may be either a **Student** or **Admin** (Generalization relationship)

ERD:



Tables converted from ERD:

ADMIN

<u>AD_ID</u>	AD_user	AD_PASSWORD
--------------	---------	-------------

QUESTIONS

<u>Q_ID</u>	Q_title	Q_a	Q_b	Q_c	Q_d	Q_correct
-------------	---------	-----	-----	-----	-----	-----------

Q_adddeddate	<u>Q_fk_ad</u>	<u>Q_fk_ex</u>
--------------	----------------	----------------

EXAMS

<u>Ex_id</u>	Ex_name
--------------	---------

SETEXAMS

<u>Set_exam_id</u>	Set_exam_date	<u>std_id_fk</u>	<u>Ex_id_fk</u>
--------------------	---------------	------------------	-----------------

SCORE

<u>score_id</u>	score	percentage	<u>Set_exam_id_fk</u>
-----------------	-------	------------	-----------------------

STUDENT

<u>Std_ID</u>	Std_name	Std_password	batchcode	<u>Std_ad_id</u>
---------------	----------	--------------	-----------	------------------

Chapter 2

Tables after 4NF:

ADMIN

<u>AD_ID</u>	AD_user	AD_PASSWORD
--------------	---------	-------------

QUESTIONS

<u>Q_ID</u>	Q_title	Q_a	Q_b	Q_c	Q_d	Q_correct
-------------	---------	-----	-----	-----	-----	-----------

Q_adddeddate	<u>Q_fk_ad</u>	<u>Q_fk_ex</u>
--------------	----------------	----------------

EXAMS

<u>Ex_id</u>	Ex_name
--------------	---------

SETEXAMS

<u>Set_exam_id</u>	Set_exam_date	<u>Std_id_fk</u>	<u>Ex_id_fk</u>
--------------------	---------------	------------------	-----------------

SCORE

<u>score_id</u>	score	percentage	<u>Set_exam_id_fk</u>
-----------------	-------	------------	-----------------------

STUDENT

<u>Std_ID</u>	Std_name	Std_password	batchcode	<u>Std_ad_id</u>
---------------	----------	--------------	-----------	------------------

FORMS:

Add New Student



Name

M.ALI

Batch Code

.....

Password

.....

Add Student

Question

what is DB?

Option A

DataBase

Option B

DataBoy

Option C

DotBase

Option D

DanishBadar

Exam

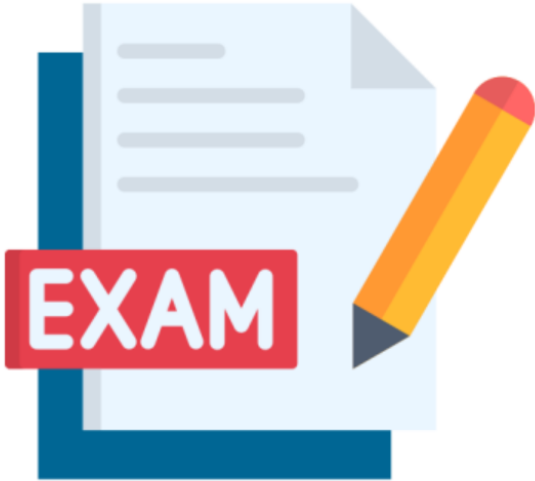
SE-A DBMS Quiz

Correct Option

DataBase

Add to Question Bank

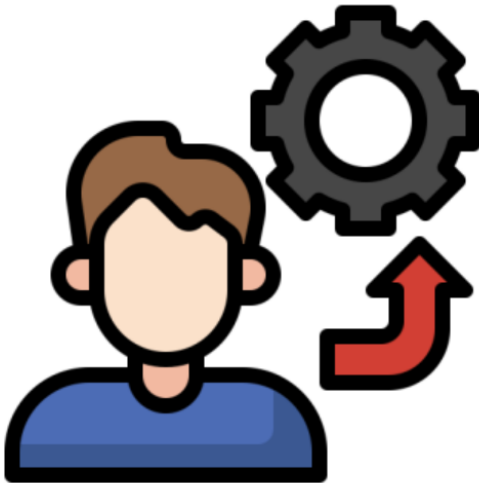
Add New Exam



Exam Name

Add New Exam

Add New Admin



User Name

Password

Add New Admin

AssignExam

Batch Code

BSDS-A

Select

Exam

CS-B OOP

Std_ID

Fatima Ma

Assign!

STUDENTS

Std_ID	Std_Name	Std_BatchCode
50	Omer Nawaz	BSDS-A
53	Kiran Farooq	BSDS-A
54	Fatima Malik	BSDS-A
56	Fatima Khan	BSDS-A
84	Hina Malik	BSDS-A
85	Ali Chaudhry	BSDS-A
86	Tariq Khan	BSDS-A
87	Kiran Nawaz	BSDS-A
101	new student	BSDS-A

EXAM ASSIGNED STUDENTS

Std_ID	Std_Name	Std_BatchCode	Exam_Name	Exam_Assign_D
24	Hina Sheikh	BSCS-B	CS-C DSA Quiz	09/18/2025
68	Ali Sheikh	BSCS-A	IT-A Cyber Quiz	09/15/2025
99	Kiran Malik	BSAI-A	CS-A C++ Quiz	10/11/2025
74	Hina Butt	BSCS-C	SE-B Web Quiz	05/20/2025
57	Kiran Khan	BSIT-B	IT-B OS Quiz	05/14/2025
73	Zain Khan	BSIT-A	DS-A Networks Quiz	02/11/2025
33	Zain Farooq	BSCS-C	DS-A Networks Quiz	10/07/2025
79	Ahmed Malik	BSCS-A	CS-C DSA Quiz	09/02/2025
92	Omer Khan	BSCS-A	IT-A Cyber Quiz	11/16/2025
84	Hina Malik	BSDS-A	CS-A C++ Quiz	06/07/2025
41	Omer Sheikh	BSIT-B	CS-C DSA Quiz	01/12/2025

See Results

Std_ID	Std_Name	Exam	Score	Percentage
1	Ahmed Yousaf	IT-A Cyber Quiz	0	0
4	Ali Butt	AI-A STAT Quiz	7	90
6	Sana Khan	IT-A Cyber Quiz	10	80
7	Sana Malik	IT-A Cyber Quiz	5	90
9	Hina Farooq	IT-A Cyber Quiz	1	10
13	Nida Yousaf	IT-B OS Quiz	9	70
15	Hassan Raza	AI-A STAT Quiz	10	50
18	Uzma Malik	AI-A STAT Quiz	8	70
21	Hina Raza	CS-C DSA Quiz	9	100
26	Tariq Butt	IT-B OS Quiz	9	90
27	Fatima Yousaf	CS-A C++ Quiz	10	60

Delete Student

Please note that Student, its related Exam Assignments and Results will be Deleted

BatchCode

BSCS-B

Student

Nida Yousaf

Select

Delete

Std_ID	Std_Name	Std_BatchCode
13	Nida Yousaf	BSCS-B
15	Hassan Raza	BSCS-B
24	Hina Sheikh	BSCS-B
25	Tariq Nawaz	BSCS-B
29	Hina Nawaz	BSCS-B
42	Hassan Khan	BSCS-B
69	Nida Farooq	BSCS-B
72	Sana Raza	BSCS-B
91	Sana Nawaz	BSCS-B
93	Zain Nawaz	BSCS-B

Delete Exam

Please note that Exam, its related Exam Assignments, Questions and Results will be Deleted

Exam

AI-A STAT Quiz

Delete

ex_id	ex_name
2	AI-A STAT Quiz
9	AI-B Python Quiz
7	CS-A C++ Quiz
1	CS-B OOP Quiz
4	CS-C DSA Quiz
6	DS-A Networks ...
10	IT-A Cyber Quiz
5	IT-B OS Quiz
3	SE-A DBMS Quiz
8	SE-B Web Quiz

Delete Score



Please note that the Score will be Permanently Deleted

Score ID

37

Delete

Score ID	Score	Student	Exam
37	0	Ahmed Yousaf	IT-A Cyber Quiz
7	7	Ali Butt	AI-A STAT Quiz
8	10	Sana Khan	IT-A Cyber Quiz
25	5	Sana Malik	IT-A Cyber Quiz
35	1	Hina Farooq	IT-A Cyber Quiz
26	9	Nida Yousaf	IT-B OS Quiz
11	10	Hassan Raza	AI-A STAT Quiz
2	8	Uzma Malik	AI-A STAT Quiz
17	9	Hina Raza	CS-C DSA Quiz

Delete Question



Please note that Question will be Deleted Permanently

Exam

CS-B OOP Quiz

Question ID

2

Select

Delete

ID	Title	Exam
2	What is encaps...	CS-B OOP Quiz
3	What does this ...	CS-B OOP Quiz
4	Which access sp...	CS-B OOP Quiz
5	Which feature a...	CS-B OOP Quiz
6	What is constru...	CS-B OOP Quiz
7	Destructor is us...	CS-B OOP Quiz
8	Which concept ...	CS-B OOP Quiz
9	What is functio...	CS-B OOP Quiz
10	Which OOP prin...	CS-B OOP Quiz
11	What is the out...	CS-B OOP Quiz

Delete Exam Assignment

Please note that the Exam Assignment and its - related Result will be Permanently Deleted



Assignment ID
1
Delete

Assignment ID	Student	Exam
20	Ahmed Yousaf	IT-A Cyber Quiz
23	Uzma Iqbal	AI-B Python Quiz
25	Ali Butt	AI-A STAT Quiz
21	Ali Farooq	IT-B OS Quiz
27	Sana Khan	IT-A Cyber Quiz
64	Sana Malik	IT-A Cyber Quiz
36	Tariq Raza	IT-B OS Quiz
34	Hina Farooq	IT-A Cyber Quiz
29	Hassan Sheikh	CS-C DSA Quiz

Chapter 3

Final Classes of the Project

- Person
- Admin
- Student
- Questions
- Exams
- Set_Exam
- Score
- ReturnClass

Their Methods and Data types:

PERSON CLASS:

- Id: String
- Name: string
- Password: string

ADMIN CLASS:

- Ad_id: **string**
- Ad_Name: **string**
- Ad_Password: **string**
- AddQuestions(): **Void**
- AddStudent(): **Void**
- SetExam(): **Void**

STUDENT CLASS:

- Std_id: **int**
- Std_Name: **String**
- Std_batchcode: **string**
- Std_Password: **String**
- TakeExam(): **Void**

QUESTIONS CLASS:

- Q_id: **int**
- Q_title: **string**
- Q_a: **string**
- Q_b: **string**

- Q_c: **string**
- Q_d: **string**
- Q_correct: **string**
- Q_addeddate: **string**
- Std_id_fk: **string**
- Exam_id_fk: **String**
- Display(): **Void**

EXAMS CLASS:

- Exam_id: **string**
- Exam_Name: **string**

SET_EXAM CLASS:

- Set_exam_id: **string**
- Set_Exam_date: **string**
- Std_id_fk: **string**
- Exam_id_fk: **string**
- setExam(): **Void**

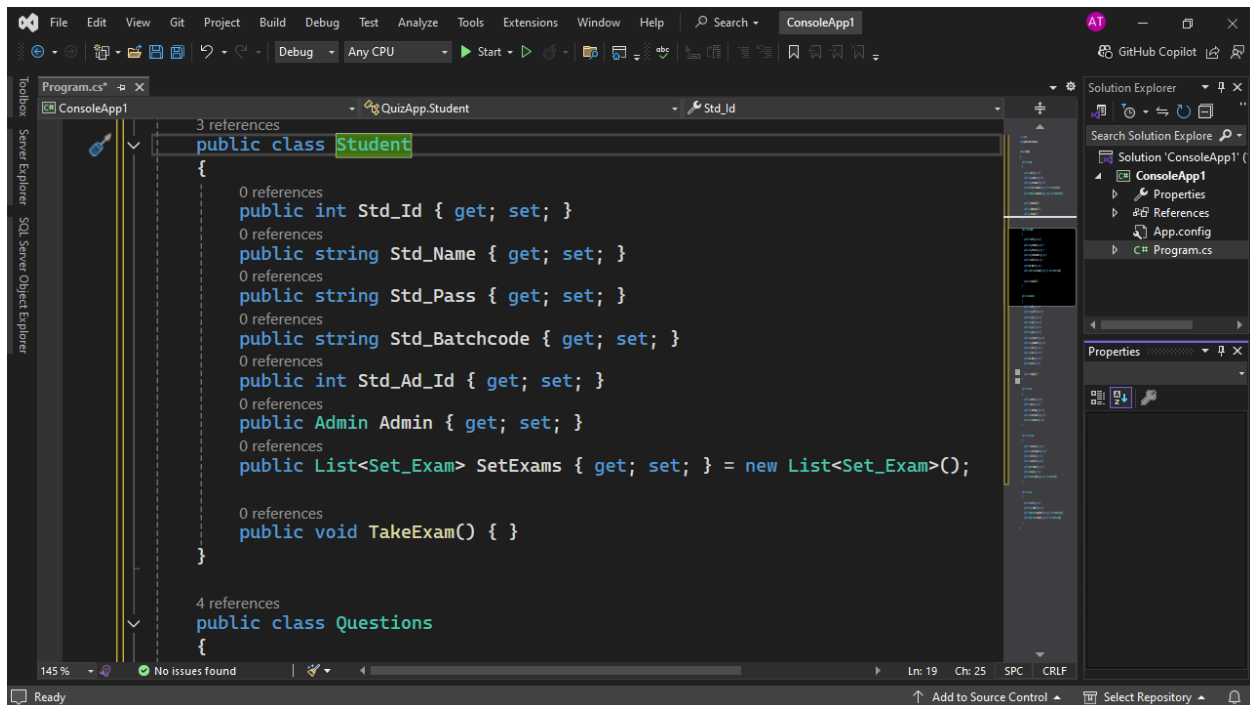
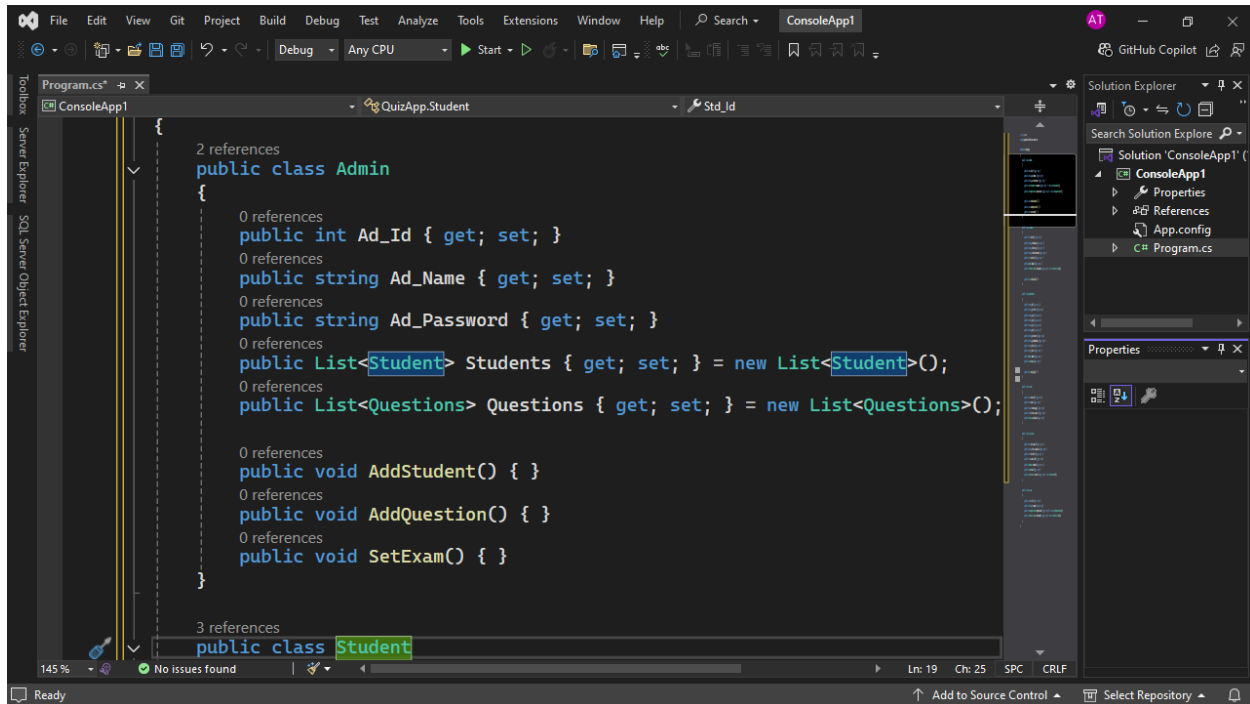
SCORE CLASS:

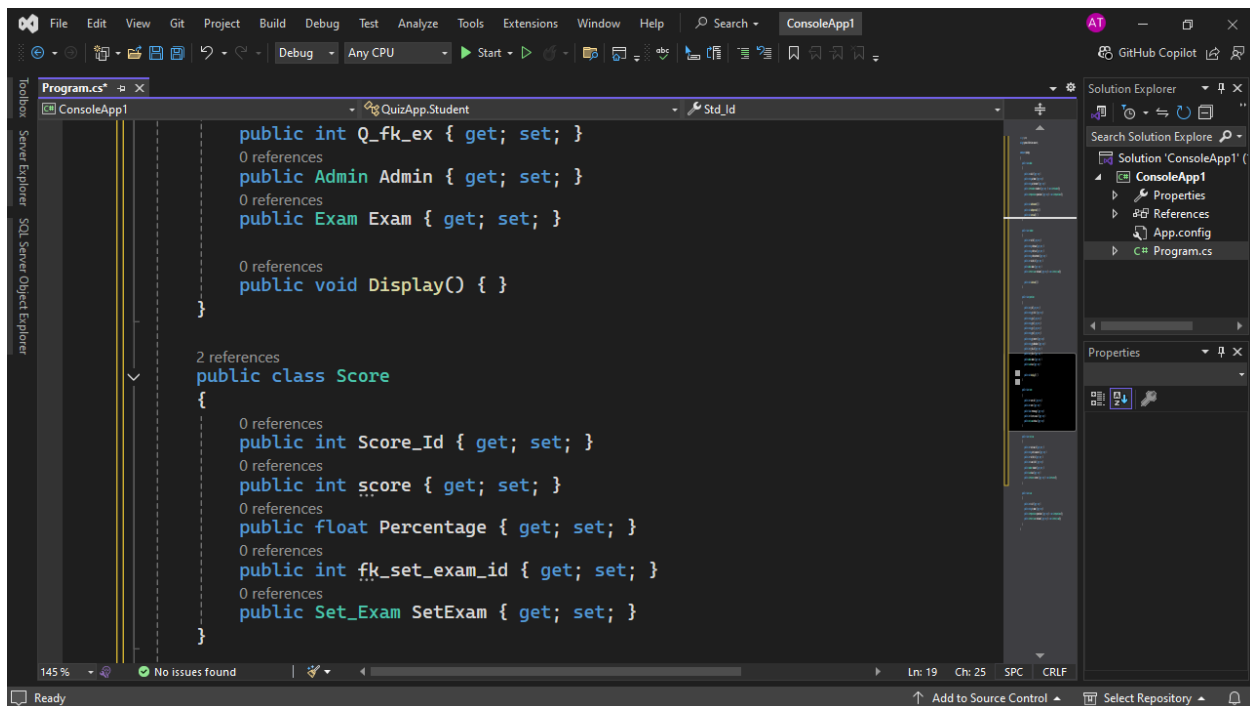
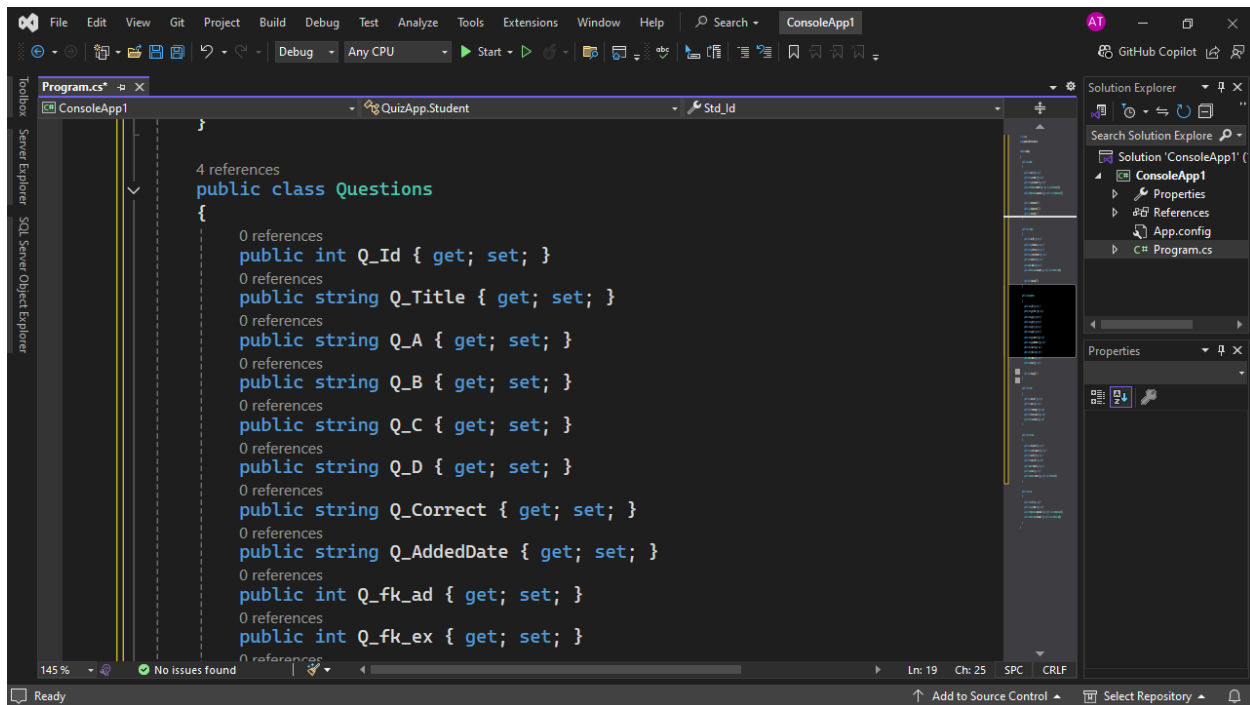
- Score_id: **String**
- Score: **string**
- Percentage: **float**
- Set_exam_id_fk: **int**
- Getscore(): **Void**

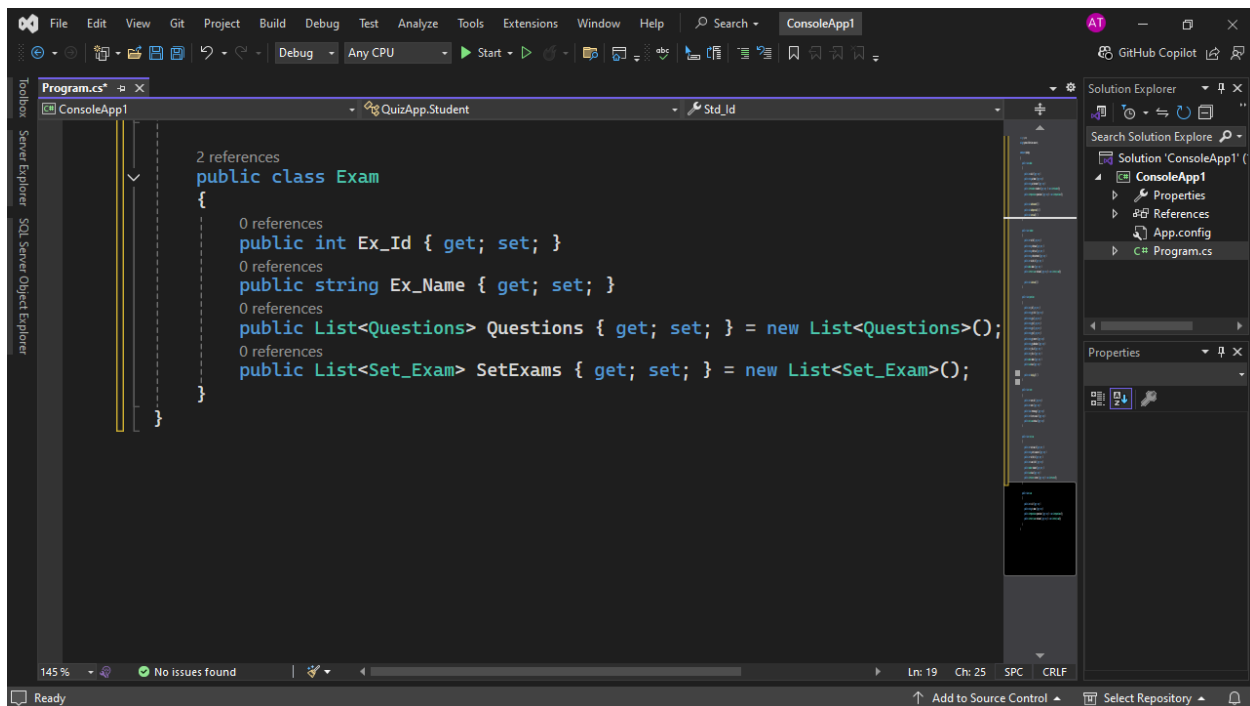
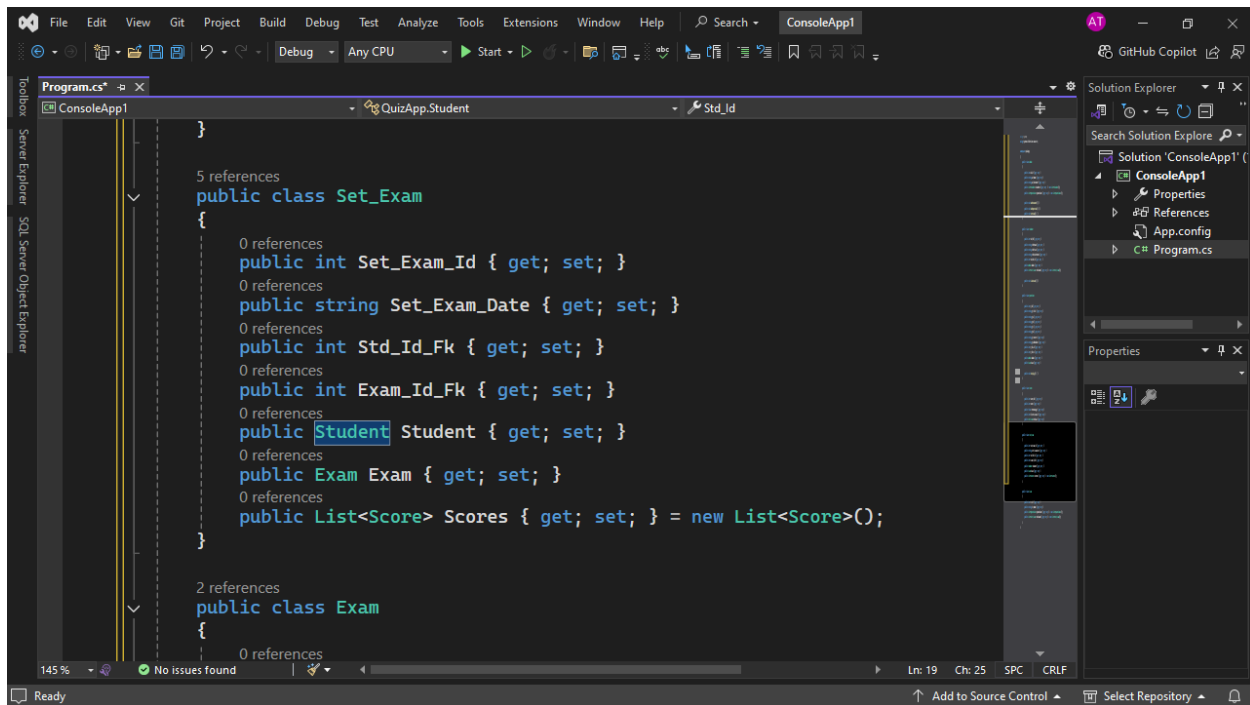
RETURN CLASS:

- Returntodb(): Void

Screen Shots From Visual Studio:







CLASS DIAGRAM:

